

WEBSAGA

Web-Based Academic ERP with Automated Question Paper Generation

In many higher educational institutions, academic administration activities such as program management, course allocation, faculty mapping, and question paper preparation are handled using manual processes or disconnected software tools. These approaches are time-consuming, error-prone, and lack proper integration between programs, branches, regulations, courses, and faculty. Additionally, question paper generation often depends on manual selection of questions, which may not ensure balanced coverage of course outcomes, Bloom's taxonomy levels, and difficulty levels. There is a need for a centralized, secure, and responsive web-based system that automates academic data management and enables systematic generation of question papers while maintaining consistency, accuracy, and compliance with academic regulations.

List of Modules:-

1. Admin

i. **Admin Dashboard**

This module should provide the key information about the system using database information like courses, programs, faculty and different module in Admin for Quick Overview, Data Visualization etc.,

ii. **Program**

This module mainly to create different **PROGRAMS** like B.tech, M.Tech, MBA etc., here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

iii. **Branch**

This module mainly to create different **BRANCHES** with the branch code like CSE, IT, CSE-AIM, CSE-AIDS etc., and branch codes like 05,12,42,45, here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

iv. **Regulation**

This module mainly to create different **REGULATIONS** like AR23,AR21,AR20 etc., here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

v. **Program – Branch Mapping (P-B Mapping)**

This module mainly to map the **BRANCHES**(which are active in Branch module) with **PROGRAMS**(which are active in Programs module) like B.tech – CSE, M.Tech - VLSI etc., here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

vi. Course

This module mainly to create different **COURSES** like Data Structures, Python Programming etc., here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

important: Course creation should map with **Branch, Regulation, Year(I,II,III,IV), Semester(I,II), Course Type(Theory, Lab, Project etc.), Elective Type(CORE, Professional Elective or Open Elective), Credits, and Course Code**

Note: in the above branch and regulation active values should take from those modules.

vii. Branch – Course Mapping(B-C Mapping)

This module mainly to map multiple **COURSES** (which are active in Course module) with **BRANCHES**(which are active in Branch module), here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

important: mapping should done with **P-B Mapping, Regulation**.

Note: in the above p-b mapping and regulation active values should take from those modules.

viii. Faculty

This module mainly to create **FACULTIES**, here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

important: Faculty creation should have **User Type** like Admin or Faculty, **Brach**, **Honorific** like Dr., Mr., Mrs. etc., **FacultyName, EMPID/PerNo., PhoneNumber, Email, and Password**.

Note: in the above branch and regulation active values should take from those modules, and here the password should be random generated password. Faculty should receive that generated password through that EmailId.

Need bulk upload of faculty with specific template

ix. Faculty – Course Mapping

This module mainly to map **FACULTY** (which are active in Faculty module) with **COURSE**(which are active in Course module), here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

important: mapping should done with **CourseType, Year, Semester, AcademicYear and ElectiveType**.

Note: in the above CourseType and Course are dependent means Course data should get after select the CourseType only and AcademicYear should be automatic and disabled field based on your system timelines means if you create a map today then that academic year should be 2025-2026 because the system timeline is between June 2025 and May 2026.

x. Course Plugins

a. Bloom's Level

This module mainly to create different **BLOOM'S LEVELS** like Remember, Understand, Apply etc., and here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

b. Difficulty Level

This module mainly to create different **DIFFICULTY LEVELS** like Easy, Moderate, Hard etc., here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

c. Unit

This module mainly to create different **UNITS** like Unit-1, Unit-1 etc., and here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

xi. QP Generation

This module mainly for generating Automated Question Paper with selecting **PROGRAM, COURSE, ASSESMENTTYPE (continuous assessment means MID-1 and MID-2, Final Exam means Regular and Supply), DateofExam, Regulation, Year, Semester and AcademicYear**.

Note: in the above all are dependent field's means:

- ✓ When select program then mapped courses with that program should fetch in course dropdown.
- ✓ After select the course from course dropdown then the field's Regulation, Year and Semester of that selected course should filled automatically.
- ✓ AcademicYear should be automatic and disabled field based on your system timelines means if you create a map today then that academic year should be 2025-2026 because the system timeline is between June 2025 and May 2026.
- ✓ Finally after done all the above then click a button to open the Question Paper template, and here you need to select CO, Bloom, Difficulty and Marks to generate random question automatically.
- ✓ After completion of the Automated QP generation, the page should provide the print option to take the print in proper PDF/Document template after clicking on **PRINT** button.
- ✓ Finally, that generated QP should save automatically at somewhere in the system after click on **SAVE** button.

2. Faculty

i. Faculty Dashboard

This module should provide the key information about the system using database information like courses, programs, faculty and different module in Admin for Quick Overview, Data Visualization etc.,

ii. My Courses

In this module the list of courses which are mapped to logged faculty should be visible here in card format. In card display **COURSECODE**, **COURSENAME**, **COURSETYPE**, **BRANCH**, **ELECTIVETYPE** and **YEAR/SEM** of that particular mapped subject and the card must have **VIEW** link to open the sub modules of that course .

a. Course Outcomes

This module mainly to create **COs** of the course and here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

b. Question

This module mainly for create **QUESTIONS**, here should perform **CRUD** operations and must provide the **status** of the each program either **active** or **inactive**.

important: Question creation should have **CO**, **BloomsLevel**, **DifficultyLevel**, **UnitNo**, **QuestionText**, **Image(if the question have images or equestions)** and **Marks**.

Note: in the above CO, bloomslevel, difficultylevel, unitno active values should take from those modules.

Need bulk upload of the questions with the specific template

iii. Change Password

In this module user can change the current password with new password.

Note:

- Both Admin and Faculty should have login and logout
- Passwords should hash before storing in database by using known familiar algorithm.
- SESSION management mandatory
- Responsive application (should works well on all type of devices like mobile, tablets, laptops and desktop screens)
- Try to develop same template for complete application